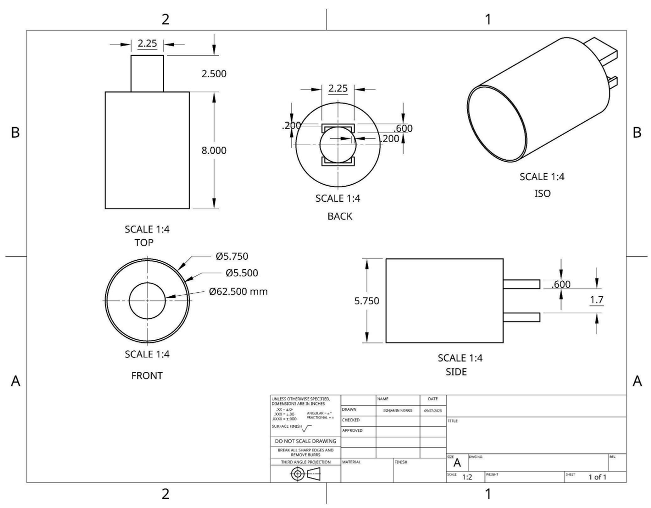
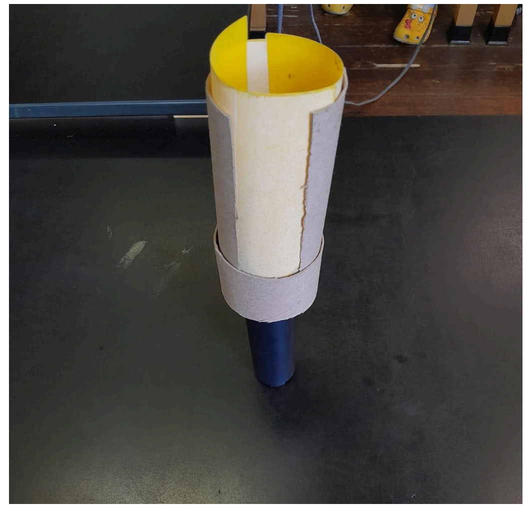
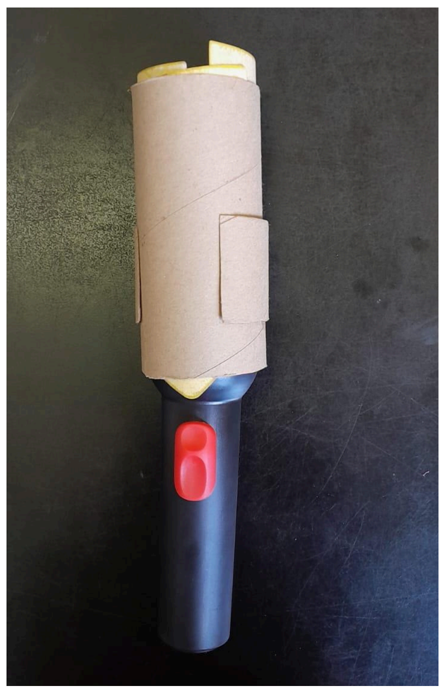
Element G: Construction of a Testable Prototype

Student E, a student in our engineering class, needs a way to get rid of wasps, hornets, and other insects that enter his room at night when he sleeps. The student's bedroom has a serious insect problem. The insects make the space noisy and affect the students' ability to rest.

The initial prototype design focused on a tube and a light source (flashlight) to attract and trap insects. Our design requirements, as seen in the initial sketches, included a mechanism to attract insects and then capture them. We thought that insects would be drawn to the light source and then become trapped on sticky paper placed inside the tube. The primary concept was that the light would lure the insects into the tube, where they would become immobilized, meeting the requirement of containing the insects in one area. The design aimed to be small and lightweight, and easy to assemble.

To create it, we take a roll of sticky paper and insert it directly into a standard paper towel tube, which measures 150.4 mm in length. For an attractant, we then insert a small flashlight approximately 50 mm into one end of the paper towel tube. The light from the flashlight draws the insects towards the sticky paper inside the tube, where they become trapped. Our evaluation indicated that this method offers a good combination of catching insects efficiently and being safe for indoor use.

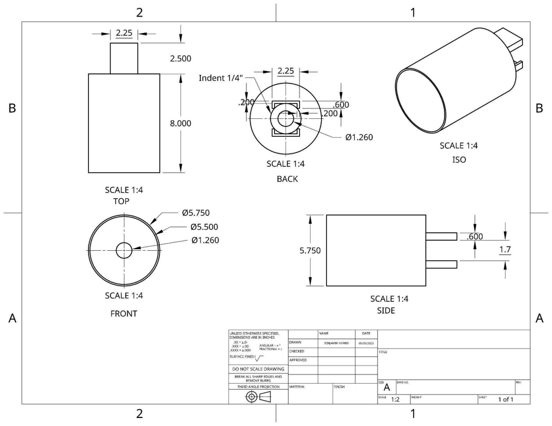


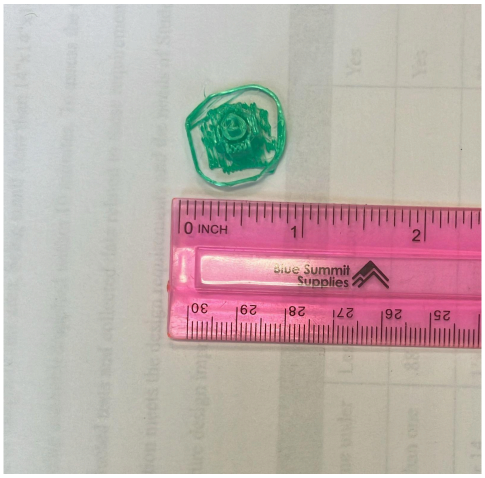



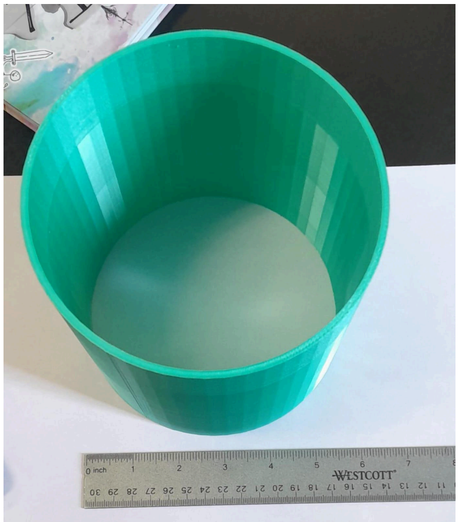
First Prototype

Based on the experts' feedback, we iterated on the first prototype to address the identified limitations and improve its effectiveness. The second prototype incorporates the following changes. We narrowed the hole where the flashlight's light shines through. We added a lens over the hole to disperse the light within the tube. This prevents the light from shining outward and attracting insects to unintended areas, and also prevents the light from blinding the insects until they are fully inside the trap, as suggested by the experts to ensure the insects are effectively contained. To address the issue of the insects not being attracted to light, as pointed out by Dr. Herbert, we will apply an attractant inside the tube to lure the insects. The overall design was improved to better contain the insects once they enter the trap, preventing escape. These changes address the experts' concerns about light effectiveness and insect containment and aim to create a more efficient and targeted insect trap. This modification also helps in making the design small and compact. The ethical consideration of not harming pets, humans, or the environment is a key factor in our design choices, particularly in the selection of the attractant.

Our testing plan covers all prioritized design requirements that can be evaluated. We will assess assembly time, which must be under 10 minutes, weight, which needs to be less than one pound, volume, which must stay under 14 cubic inches, and effectiveness in neutralizing or capturing insects in a specific area. However, we were unable to test whether the design is ethical, meaning it does not harm humans, pets, or the environment, or whether it has an appealing aesthetic. However, we have confirmed the design criteria by researching the materials and interviewing Student E. Our tests will provide us data to confirm we meet our four highest priority design criteria, such as an assembly time under ten minutes by testing for an average assembly time, a volume less than 14 cubic inches by measuring the volume of the prototype, a weight less than one pound by measuring mass of the prototype, and neutralization of insects in one spot by counting the amount of insects trapped in different locations.







Second Prototype

The second prototype, incorporating the design changes, will be constructed and tested outside of Student E's house. Testing will evaluate the effectiveness of the attractant and light dispersion in capturing and containing insects. We expect the modifications to result in a more effective and reliable insect trap, adhering to the design requirements. The second prototype was printed at a smaller scale than expected, making it unusable for the project. When we attempted another print, it came out too large, which also rendered it unsuitable.